Dongho Lee

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Research Interests

Keywords : Symplectic Geometry and Dynamics, Three-body Problem

My primary research focuses on understanding dynamical systems through the framework of symplectic geometry. I am particularly interested in classical Hamiltonian systems such as the three-body problem, where families of periodic orbits exhibit rich bifurcation phenomena. Using tools from Floer theory, I investigate the geometric and topological mechanisms underlying these behaviors, and I am also interested in connecting these theoretical insights to practical applications in celestial mechanics and related areas.

Education

Ph.D. in Mathematics
Department of Mathematical Science, Seoul National University (SNU) Mar 2018 – Feb 2025

Bachelor of Science in Mathematics
Bachelor of Economics in Economics, Minor in Physics
College of Liberal Studies, SNU Mar 2013 – Feb 2018
Graduated with Honors

Research Experience

Postdoctoral Researcher

Center for Quantum Structures in Modules and Spaces (QSMS), SNU May 2025 – Present

Research Institute of Mathematics, SNU Mar 2025 – Apr 2025
Postdoctoral Supervisor: Prof. Cheol-hyun Cho

Graduate Student Researcher

Department of Mathematical Sciences, SNU Mar 2018 – Feb 2025
Ph.D Advisor: Prof. Otto van Koert

Publications

1. **Conley-Zehnder Indices of Spatial Rotating Kepler Problem** arXiv Preprint
2. **Global Hypersurfaces of Section and the Spatial Kepler Problem**
   Advisor: Prof. Otto van Koert Ph.D. Thesis, Dec 2024
3. **Global Hypersurfaces of Section for Geodesic Flows on Convex Hypersurfaces**
   with Sunghae Cho *Archiv der Mathematik*, 31 Jul 2024

Talks and Presentations

Sildes and posters are available at my homepage.

Invited Talks

1. **Generic Bifurcation of Hamiltonian Orbits**
   Symplectic Learning Seminar, SNU, Seoul, Korea — 12 Dec 2025
2. **Hamiltonian Dynamics and Three-body Problem**
   Rookie's Pitch, SNU, Seoul, Korea — 14 Oct 2025
3. **Three-body Problem and Related Problems**
   QSMS 2025 Summer Workshop, Goseong, Korea — 20–23 Aug 2025
4. **Periodic Orbits of Rotating Kepler Problem**
   Billiards and Quantitative Symplectic Geometry, Heldelberg, Germany — 14 – 18 Jul 2025
5. **Closed orbits of the spatial rotating Kepler problem**
   2025 Korean Mathematical Society Spring Meeting, Daejeon, Korea — 24 – 26 Apr 2025

Poster Talks

1. **Periodic Orbits and Symplectic Invariants in the Rotating Kepler Problem**
   Celestial Mechanics & N-body Dynamics, Tokyo, Japan — 15–16 Nov 2025

Conference Participation

New Developments in Symplectic Geometry — 25 – 28 Nov 2025
Seoul National University, Seoul, Korea

East Asian Symplectic Conference 2025 — 25 – 29 Sep 2025
Hokkaido University, Sapporo, Japan

2025 Symplectic Retreat — 23 – 26 Jan 2025
Stanford Hotel, Tongyeong, Korea

From Hamiltonian Dynamics to Symplectic Topology and Beyond — 3 – 7 Jun 2024
Institut Henri Poincaré, Paris, France

Topology of 4-manifolds and Related Topics — 21 – 27 Jan 2024
Ocean Suites Hotel, Jeju, Korea

East Asian Symplectic Conference 2023 — 29 Oct – 4 Nov 2023
Ocean Suites Hotel, Jeju, Korea

QSMS Workshop on Symplectic Geometry and Related Topics — 5 – 10 Feb 2023
Ocean Suites Hotel, Jeju, Korea

Workshop on Symplectic Dynamics — 27 Jun – 1 Jul 2022
Instituto Superior Técnico, Lisboa, Portugal

Symplectic Geometry and Beyond (Part I) — 7 – 10 Feb 2022
Ocean Suite Hotel, Jeju, Korea

Fukaya 60 Geometry and Everything — 17 – 22 Feb 2019
Kyoto University, Kyoto, Japan

Personal Affiliations

Korean Mathematical Society, Korea 2025 – Present

QSMS-BK21 Symplectic Seminar, Korea 2021 – Present

Teaching Experience

Teaching and Course Assistant, SNU Mar 2018 – Feb 2025

Major Courses

- **Algebraic Topology** (Graduate Course) Fall 2019
- **Introduction to Topology** Spring 2023, Spring 2020
- **Introduction to Differential Geometry** Spring, Fall 2021
- **Selected Topics Seminar** (College of Liberal Studies) Spring 2023

General Education Courses

- **Head Teaching Assistant for Introductory Mathematics Courses** Spring 2020
- **Engineering Mathematics** Fall 2024, Fall, Summer, Spring 2023, Fall 2020
- **Calculus for Life Science** Fall 2022, Spring 2021
- **Fundamentals and Applications of Mathematics** Spring 2022
- **Calculus Practice** Fall, Spring 2018
- **Calculus** Fall, Spring 2018

Awards and Honors

Merit-based Scholarship, Department of Mathematical Sciences, SNU 2018

National Scholarship, Korean Student Aid Foundation 2015

Merit-based Scholarship, College of Liberal Studies, SNU 2013

Languages

Korean: Native

English: Fluent

Japanese, French: Basic – Intermediate

References

Prof. Otto van Koert
Department of Mathematical Sciences, SNU
`okoert@snu.ac.kr`
Relationship: Ph.D. Advisor

Prof. Cheol-hyun Cho
Department of Mathematics, POSTECH
`chocheol@postech.ac.kr`
Relationship: Postdoctoral Supervisor